1) Explain the behaviour of HUB device.

- Hub is a layer 1 device.
- Hub broadcasts data to all.
- The mother board circuit connected in bus 'topology' - hence if data transmittes or Sends from both direction at Same time, there is chance for data ~~collion~~ collision and lossing of data.
- data is considered as signals.
- here the data transmission is in half duplex.

* Due to data is broadcasting traffic will be in network.

2) Explain the behaviour of switc device.

→ • It is a layer 2 device
works in data link layer
• Also switch is a intelligent device
• Swith maintain a table called MAC table and the learning is called 'address learnig'.
* Switch Sends data in both broadcast and unicast method.
• The switch will save source MAC and source port in MAC table with a 5 minuts time validity.
• Used to conned ~~same network~~
• In switch connectivity there is a broadcasting issue called single broadcast domain. ie a data is send to one device and the switch will broadcast it to all.

3)

→

- If a data come to switch, the switch will check the Mac table for Source port and destination mac.
  - if present it will forward unicast
  - If not present it will broadcast to all.

3) Explain the behaviour of routers,

→ • It is a layer 3 device. ie works in Network layer.

• Used to connect devices in different network.

• ~~b~~ It breaks the concept of single broad cast domain.

• It maintain a table called routing table, which contain source port, Network ID, prefix

• Usually the connected ports assign first IP of the network and called "directly connected route".

• If a data ~~came~~ come to router, the device check the routing table for ~~if~~ the directly connected routes is present or not the forward to accordingly and if no route, it will drop.